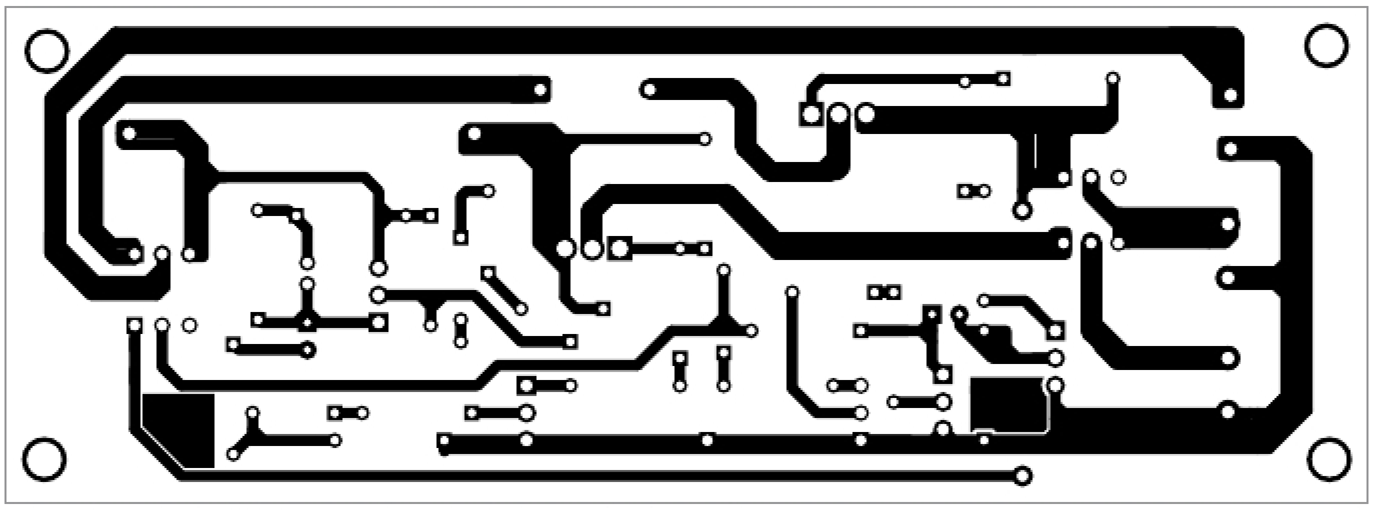

Fig. 5: Actual-size PCB layout for solar day lamp with battery backup

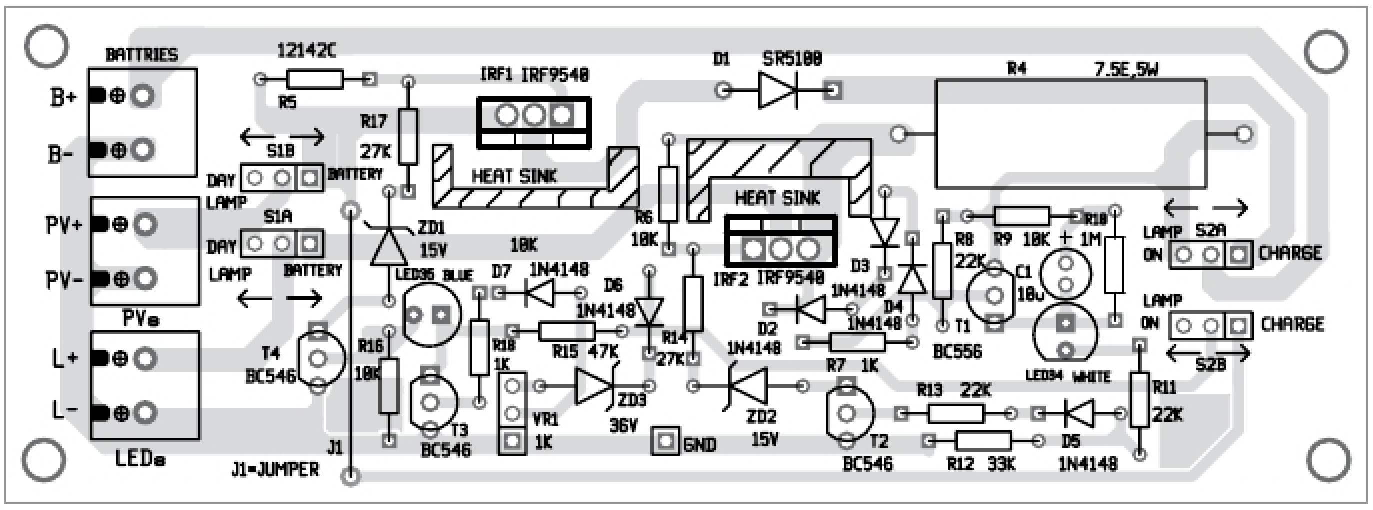

Fig. 6: Components layout of the PCB

Fig. 7: Two 10Wp PV panels (Size: 600mm×350mm)

be slightly tweaked to get optimum performance.

Three lead-acid batteries (BATT.1, BATT.2, and BATT.3) are connected in series. Fuse F1 is connected between BATT.2 and BATT.3. The author's prototype in Fig.1 shows the battery bank (BATT.1 through BATT.3) mounted on a single base. A small PCB on top of the battery bank is used to mount the fuse.

The proposed circuit operates in the following three different modes:

1. Day lamp mode
2. Battery charging mode
3. Lamp on mode (using battery power)

Two double-pole double-throw (DPDT) switches S1 and S2 are used to change the operating modes of the circuit. IRF1 MOSFET is used to turn off the circuit when batteries are fully charged, and IRF2 MOSFET is used to turn off the circuit when batteries are fully discharged.

Day lamp mode. In this mode, the PV voltage generated by panels is directly connected to the lamp. This is done by moving switch S1 to 'day lamp' position. LED arrays L1, L2, and L3 turn on, and no other part of the circuit gets power supply. The design details of the lamp are shown in Box 2.

Battery charging mode. For charging the battery, S1 is moved to 'battery' position and S2 to 'charge' position. The glowing of LED35 indicates battery charging mode. In this mode, panel current passes through IRF1, D1, and the battery bank.

In order to ensure IRF1 is on, the circuit comprising D6, ZD3, and VR1 is connected to PV+ through S2B and S1B. If the battery voltage is less than 36.7V, T3 is cut off. Current flowing through resistor R15 and LED35 turns on T4. This pulls down the gate voltage of IRF1 with respect to the source terminal. Hence, IRF1 turns on.

When batteries are fully charged, the voltage across VR1 turns on T3, and IRF1 is turned off. VR1 is used for setting the voltage for charging batteries. The purpose of diode D1 is to stop reverse battery current when PV voltage is not available. The solar panels can generate an open-circuit voltage of about 40V. Hence, it can charge the batteries up to 39V.

Lamp on mode. In this mode, the lamp is turned on using power from the battery bank (BATT.1 through BATT.3) When S2 is moved to 'lamp on' position, the battery bank gets connected to the lamp through resistor R4 and IRF2. LED34 glows, indicating that the lamp is on. When the battery voltage goes down, the drop across R4 reduces, causing IRF2 to turn off.

Construction and testing

An actual-size PCB layout for the solar day lamp with battery backup is shown in Fig. 5 and its components layout in Fig. 6. After assembling the circuit on PCB, connect PV panels, LED arrays, and batteries.

Fig. 7 shows two PV panels mounted on aluminium frames. PV panels should be placed where sunlight is available throughout the day. Usually, the wire/cable from the PV panel to the circuit is long, so a good-quality 1 sq mm cable should be used. But if the wire from PCB to the lamp is short (less than 3 metres), an ordinary wire can be used.

In the evening before leaving office and in weekends, set to battery charging mode.

Note. Instead of using individual 1W LEDs, you can use metal-clad LED PCBs (either round or strip form) that are normally used in LED lamps and tubelights. EFY

Dr Vijay Deshpande currently works for solar energy projects. He has also worked as electronics hardware engineer in several companies